5.2 RESULTS

Figure 3 shows MIN-K% PROB achieves an AUC of 0.88, outperforming baselines in detecting copyrighted books. We apply the optimal threshold of MIN-K% PROB to the test set of 10,000 snippets from 100 books from Books3. Table 2 represents the top 20 books with the highest predicted contamination rates. Figure 4 reveals nearly 90% of the books have an alarming contamination rate over 50%.

Method	Book
Neighbor	0.75
PPL	0.84
Zlib	0.81
Lowercase	0.80
MIN-K% PROB	**0.88**

Figure 3: AUC scores for detecting the validation set of copyrighted books on GPT-3.

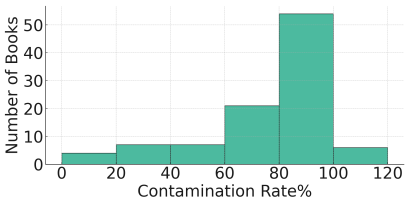


Figure 4: Distribution of detected contamination rate of 100 copyrighted books.

Table 2: Top 20 copyrighted books in GPT-3's pretraining data. The listed contamination rate represents the percentage of text excerpts from each book identified in the pretraining data.

Contamination %	Book Title	Author	Year
100	The Violin of Auschwitz	Maria Àngels Anglada	2010
100	North American Stadiums	Grady Chambers	2018
100	White Chappell Scarlet Tracings	Iain Sinclair	1987
100	Lost and Found	Alan Dean	2001
100	A Different City	Tanith Lee	2015
100	Our Lady of the Forest	David Guterson	2003
100	The Expelled	Mois Benarroch	2013
99	Blood Cursed	Archer Alex	2013
99	Genesis Code: A Thriller of the Near Future	Jamie Metzl	2014
99	The Sleepwalker's Guide to Dancing	Mira Jacob	2014
99	The Harlan Ellison Hornbook	Harlan Ellison	1990
99	The Book of Freedom	Paul Selig	2018
99	Three Strong Women	Marie NDiaye	2009
99	The Leadership Mind Switch: Rethinking How We Lead in the New World of Work	D. A. Benton, Kylie Wright-Ford	2017
99	Gold	Chris Cleave	2012
99	The Tower	Simon Clark	2005
98	Amazon	Bruce Parry	2009
98	Ain't It Time We Said Goodbye: The Rolling Stones on the Road to Exile	Robert Greenfield	2014
98	Page One	David Folkenflik	2011
98	Road of Bones: The Siege of Kohima 1944	Fergal Keane	2010

6 CASE STUDY: DETECTING DOWNSTREAM DATASET CONTAMINATION

Assessing the leakage of downstream task data into pretraining corpora is an important issue, but it is challenging to address given the lack of access to pretraining datasets. In this section, we investigate the possibility of using MIN-K% PROB to detect information leakage and perform ablation studies to understand how various training factors impact detection difficulty. Specifically, we continually pretrain the 7B parameter LLaMA model (Touvron et al., 2023a) on pretraining data that have been purposefully contaminated with examples from the downstream task.

6.1 EXPERIMENTS

Experimental setup. To simulate downstream task contamination that could occur in real-world settings, we create contaminated pretraining data by inserting examples from downstream tasks into a pretraining corpus. Specifically, we sample text from the RedPajama corpus (TogetherCompute, 2023) and insert formatted examples from the downstream datasets BoolQ (Clark et al., 2019), IMDB (Maas et al., 2011), Truthful QA (Lin et al., 2021), and Commonsense QA (Talmor et al., 2019) in contiguous segments at random positions in the uncontaminated text. We insert 200 (positive) examples from each of these datasets into the pretraining data while also isolating a set of 200 (negative) examples from

each dataset that are known to be absent from the contaminated corpus. This creates a contaminated pretraining dataset containing 27 million tokens with 0.1% drawn from downstream datasets.

We evaluate the effectiveness of MIN-K% PROB at detecting leaked benchmark examples by computing AUC scores over these 400 examples on a LLaMA 7B model finetuned for one epoch on our contaminated pretraining data at a constant learning rate of 1e-4.

Main results. We present the main attack results in Table 3. We find that MIN-K% PROB outperforms all baselines. We report TPR@5%FPR in Table 7 in Appendix A, where MIN-K% PROB shows 12.2% improvement over the best baseline.

Table 3: AUC scores for detecting contaminant downstream examples. **Bold** shows the best AUC score within each column.

Method	BoolQ	Commonsense QA	IMDB	Truthful QA	Avg.
Neighbor	0.68	0.56	0.80	0.59	0.66
Zlib	0.76	0.63	0.71	0.63	0.68
Lowercase	0.74	0.61	0.79	0.56	0.68
PPL	0.89	0.78	0.97	0.71	0.84
MIN-K% PROB	**0.91**	**0.80**	**0.98**	**0.74**	**0.86**

6.2 RESULTS AND ANALYSIS

The simulation with contaminated datasets allows us to perform ablation studies to empirically analyze the effects of *dataset size*, *frequency of data occurrence*, and *learning rate* on detection difficulty, as theorized in section 2.1. The empirical results largely align with and validate the theoretical framework proposed. In summary, we find that detection becomes more challenging as data occurrence and learning rate decreases, and the effect of dataset size on detection difficulty depends on whether the contaminants are outliers relative to the distribution of the pretraining data.

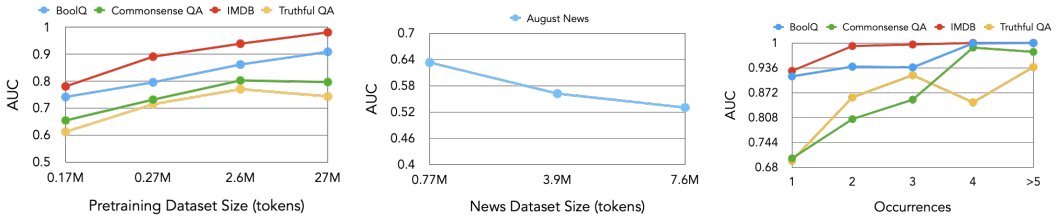
Pretraining dataset size. We construct contaminated datasets of 0.17M, 0.27M, 2.6M and 26M tokens by mixing fixed downstream examples (200 examples per downstream task) with varying amounts of RedPajama data, mimicking real-world pretraining. Despite the theory suggesting greater difficulty with more pretraining data, Figure 5a shows AUC scores counterintuitively increase with pre-training dataset size. This aligns with findings that LMs better memorize tail outliers (Feldman, 2020; Zhang et al., 2021). With more RedPajama tokens in the constructed dataset, downstream examples become more significant outliers. We hypothesize that their enhanced memorization likely enables easier detection with perplexity-based metrics.

To verify the our hypothesis, we construct control data where contaminants are not outliers. We sample Real Time Data News August 2023[4], containing post-2023 news absent from LLaMA pretraining. We create three synthetic corpora by concatenating 1000, 5000 and 10000 examples from this corpus, hence creating corpora of sizes 0.77M, 3.9M and 7.6M tokens respecitvely. In each setting, we consider 100 of these examples to be contaminant (positive) examples and set aside another set of 100 examples from News August 2023 (negative). Figure 5b shows AUC scores decrease as the dataset size increases.

Detection of outlier contaminants like downstream examples gets easier as data size increases, since models effectively memorize long-tail samples. However, detecting general in-distribution samples from the pretraining data distribution gets harder with more data, following theoretical expectations.

Data occurrence. To study the relationship between detection difficulty and data occurrence, we construct a contaminated pretraining corpus by inserting multiple copies of each downstream data point into a pre-training corpus, where the occurrence of each example follows a Poisson distribution. We measure the relationship between the frequency of the example in the pretraining data and its AUC scores. Figure 5c shows that AUC scores positively correlates with the occurrence of examples.

[4] https://huggingface.co/datasets/RealTimeData/News_August_2023



(a) Outlier contaminants, e.g., downstream examples, become easier to detect as dataset size increases.

(b) In-distribution contaminants, e.g., news articles, are harder to detect as dataset size increases.

(c) Contaminants that occur more frequently in the dataset are easier to detect.

Figure 5: We show the effect of contamination rate (expressed as a percentage of the total number of pretraining tokens) and occurrence frequency on the ease of detection of data contaminants using MIN-K% PROB.

Learning rate. We also study the effect of varying the learning rates used during pretraining on the detection statistics of the contaminant examples (see Table 4). We find that raising the learning rate from 10^{-5} to 10^{-4} increases AUC scores significantly in all the downstream tasks, implying that higher learning rates cause models to memorize their pretraining data more strongly. A more in-depth analysis in Table 8 in Appendix A demonstrates that a higher learning rate leads to more memorization rather than generalization for these downstream tasks.

Table 4: AUC scores for detecting contaminant downstream examples using two different learning rates. Detection becomes easier when higher learning rates are used during training. **Bold** shows the best AUC score within each column.

Learning rate	BoolQ	Commonsense QA	IMDB	LSAT QA	Truthful QA
1×10^{-5}	0.64	0.59	0.76	0.72	0.56
1×10^{-4}	**0.91**	**0.80**	**0.98**	**0.82**	**0.74**

7 CASE STUDY: PRIVACY AUDITING OF MACHINE UNLEARNING

We also demonstrate that our proposed technique can effectively address the need for auditing machine unlearning, ensuring compliance with privacy regulations (Figure 6).

7.1 BACKGROUNDING

The right to be forgotten and machine unlearning. In today's landscape of machine learning systems, it is imperative to uphold individuals' "right to be forgotten", a legal obligation outlined in regulations such as the General Data Protection Regulation (GDPR) (Voigt & Von dem Bussche, 2017) and the California Consumer Privacy Act (CCPA) (Legislature, 2018). This requirement allows users to request the removal of their data from trained models. To address this need, the concept of machine unlearning has emerged as a solution for purging data from machine learning models, and various machine unlearning methods have been introduced (Ginart et al., 2019; Liu et al., 2020; Wu et al., 2020; Bourtoule et al., 2021; Izzo et al., 2021; Sekhari et al., 2021; Gupta et al., 2021; Ye et al., 2022).

Recently, Eldan & Russinovich (2023) introduced a novel approach for performing machine unlearning on LLMs. This approach involves further fine-tuning the LLMs with alternative labels for specific tokens, effectively creating a modified version of the model that no longer contains the to-be-unlearned content. Specifically, the authors demonstrated the efficacy of this method using the `LLaMA2-7B-chat` model (Touvron et al., 2023b), showcasing its ability to "unlearn" information from the Harry Potter book series which results in the `LLaMA2-7B-WhoIsHarryPotter` model[5]. In this case study, we aim to assess whether this model successfully eliminates memorized content related to the Harry Potter series.

[5]Available at `https://huggingface.co/microsoft/Llama2-7b-WhoIsHarryPotter`.